

Problem 15.24

Which of the following does not represent a definition of $a_{\overline{n}}$?

(a) $v^n \left(\frac{(1+i)^n - 1}{i} \right)$

(b) $\frac{1 - v^n}{i}$

(c) $v + v^2 + \cdots + v^n$

(d) $v \left(\frac{1 - v^n}{1 - v} \right)$

(e) $\frac{s_{\overline{n}}}{(1+i)^{n-1}}$

Solution

Recall that

$$a_{\overline{n}} = \frac{1 - v^n}{i} = v + v^2 + \cdots + v^n.$$

Also,

$$s_{\overline{n}} = a_{\overline{n}}(1+i)^n.$$

Option (a)

Since

$$s_{\overline{n}} = \frac{(1+i)^n - 1}{i},$$

we have

$$v^n \left(\frac{(1+i)^n - 1}{i} \right) = \frac{s_{\overline{n}}}{(1+i)^n} = a_{\overline{n}}.$$

So option (a) is a valid representation.

Option (b)

By definition,

$$a_{\overline{n}} = \frac{1 - v^n}{i}.$$

So option (b) is valid.

Option (c)

The present value of payments of 1 at times $1, 2, \dots, n$ is

$$v + v^2 + \dots + v^n = a_{\overline{n}|}.$$

So option (c) is valid.

Option (d)

Using the geometric series formula,

$$v + v^2 + \dots + v^n = v \left(\frac{1 - v^n}{1 - v} \right).$$

So option (d) is valid.

Option (e)

Since

$$s_{\overline{n}|} = a_{\overline{n}|}(1 + i)^n,$$

it follows that

$$a_{\overline{n}|} = \frac{s_{\overline{n}|}}{(1 + i)^n},$$

not

$$\frac{s_{\overline{n}|}}{(1 + i)^{n-1}}.$$

Therefore, option (e) does not represent $a_{\overline{n}|}$.

Hence, the answer is

$$\boxed{\text{(e)}}.$$